Forward, like Dyson, imagined his scheme practical in the twenty-second century. When I look at the technical challenges, I think longer.

Gravitational Slingshots in a Black-Hole Binary



My third example is my own wild—very wild!—variant of an idea due to Dyson (1963).

Suppose you want to fly across much of the universe (not just interstellar travel, but intergalactic travel) at near light speed in a few years of your own life. You can do so with the aid of two black holes that are orbiting each other, a *black-hole binary*. They must be in a highly elliptical orbit and must be large enough that their tidal forces do not destroy your ship.

Using chemical or nuclear fuel, you navigate your ship into an orbit that comes close to one of the black holes: a so-called zoom-whirl orbit (Figure 13.4). Your ship zooms close to the hole, whirls around it a few times, and then, when the hole is traveling nearly directly toward its companion, the ship zooms out, crosses over to the companion hole, and slides into a whirl around it. If the two holes are still headed toward each other, the whirl is brief: you zoom back toward the first hole. If the holes are no longer headed toward each other, the whirl is much longer; you must park yourself in orbit around the second hole until the holes are again headed toward each other, and then launch back toward the first hole. In this way, always traveling between holes only when the holes are approaching each other, your ship gets boosted to higher and higher speeds, approaching as close as you wish to the speed of light if the binary is sufficiently elliptical.

It is a remarkable fact that you only need a small amount of rocket fuel to control how long you linger around each hole. The key is to